Literature Review

NBA Shot Quality Capstone

DAT 490 — Data Science Capstone (2026 Summer A)

NBA Analytics Team
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Review of the Literature

Our project sits at an important position between three threads in basketball analytics research: the broader rise of quantitative decision making in the NBA, the modeling of shot quality and expected possession value, and the use of those models to evaluate individual players. This review walks through each of those threads in turn, starting with the analytics era as a whole, narrowing into the theoretical and modeling work that defines shot quality, and ending with the modern player evaluation literature and the ethical considerations that come with it. The goal is to position our own work in a field that has matured rapidly over the last two decades and is still actively producing new methodology today.

The Analytics Era in the NBA

Quantitative analysis of basketball games dates back to the late 1940s (Poropudas & Halme, 2023), but the modern era of basketball analytics is widely understood to have started in the early 2000s with the publication of Dean Oliver's *Basketball on Paper* (Oliver, 2004). Oliver argued for replacing simple per-game averages with tempo-adjusted metrics such as points per possession, and he proposed the “four factors” framework — effective field goal percentage, free throw rate, offensive rebounding rate, and turnover rate — as the four independent dimensions that drive team success. Subsequent work by John Hollinger and David Berri extended this approach to composite measures of individual player value and to the connections between productivity, salaries, and team winning percentage (Poropudas & Halme, 2023). Poropudas and Halme (2023) themselves revisit Oliver's framework in light of the last two decades of league play and show that the relationship between the four factors and team efficiency ratings is non-linear, with the marginal impact of any one factor depending on the values of the others.

The infrastructure underneath these models has changed almost as much as the models themselves. Box-score summaries gave way to possession-by-possession, play-by-play data in the late 1990s, and the NBA installed optical player and ball tracking in every arena starting in the 2009–10 season (Poropudas & Halme, 2023). That tracking data is what makes the modern shot-quality tracking possible, and it is the same data feeding our own analysis. Combined with shifts like the “three-point revolution” — the league-wide move away from mid-range shots toward layups and threes — the analytics era has reshaped how teams build rosters, allocate possessions, and evaluate the players who take their shots.

Shot Selection as an Optimization Problem

One of the earliest theoretical contributions in this space comes from Skinner (2011), who framed shot selection as an optimization problem rather than an efficiency calculation. Skinner argued that the value of any given shot is not just its own expected points, but its expected points relative to the shot the offense could have generated by passing it up. Under this view, a player who only takes shots with very high expected value is not necessarily an efficient scorer, because they may be passing up enough other reasonable shots that the team's overall scoring rate suffers. Skinner's framing introduced the vocabulary of marginal value, opportunity cost, and expected outcome that the rest of the shot-quality framework builds on, and it is the conceptual model of every “expected” metric we use in our own work.

Modeling Shot Quality and Expected Possession Value

A highly influential body of modeling work in this area comes from Cervone, D'Amour, Bornn, and Goldsberry (2014), whose POINTWISE framework introduced the concept of *expected possession value* (EPV). Rather than evaluating possessions only at the moment a shot is taken, POINTWISE models the value of a possession at every moment — every pass, every dribble, every off-ball cut — as a function of the current spatial configuration of the players and the ball. The accompanying technical paper develops the multiresolution stochastic process model that makes those moment-by-moment estimates numeric (Cervone, D'Amour, Bornn, & Goldsberry, 2014). Together, these two papers established that high-frequency optical tracking data could be used not just to describe what happened on a possession, but to estimate what *should have* happened given the choices available to the offense. That distinction informs the methodological foundation of our shot-success model.

Modern Player Evaluation Through Shot Quality

More recent work has applied the EPV lineage to specific questions about individual player value. A 2024 paper introduces *Expected Points Above Average* (EPAA), a Bayesian hierarchical model that compares a player's scoring contributions against what an average player would have produced from the same shots, giving a per-player measure of shot-quality-adjusted performance (Expected Points Above Average, 2024). The approach is a direct representation of what we propose for RQ1 — taking residuals between actual makes and model-expected makes, then using those residuals to surface players who consistently outperform expectations.

Kono and Fujii (2024) extend player evaluation metrics in a different direction by modeling off-ball scoring opportunities rather than just the shooter's expected outcome. Their BMOS (Ball Movement for Off-ball Scoring) and BIMOS (Ball Intercept and Movement for Off-ball Scoring) models use STATS SportVU tracking data from 630 NBA games in the 2015–16 regular season to estimate the probability that an off-ball player will end up in a position to score, accounting for passing, dribbling, and the chance of interception along the way. The work makes a useful theoretical point that although data-driven EPV models are powerful, theory-based models grounded in physical parameters of the players and the ball can offer more interpretable results, which matters when the goal is evaluating decisions rather than just predicting outcomes.

The methodological foundations of residualized player metrics have also received recent attention. Bajons and Kook (2025), working primarily in soccer, present a general framework for player evaluation metrics built on differences between actual and expected outcomes. They show that the commonly used "above expectation" metrics — Goals Above Expectation in soccer, and by extension our shot-quality residuals in basketball — are mathematically related to Rao's score test in parametric regression models, and they propose residualized versions of those metrics that forms valid inference even when the underlying expected-outcome model is a flexible machine learning estimator. Their framework directly addresses one of the open methodological questions in our own RQ1: how confident can we be that a player's positive residual reflects real skill rather than noise? Their answer is the kind of refinement we may incorporate as we develop our analysis.

Ethics and Fairness in Algorithmic Player Evaluation

The shot-quality and player evaluation models discussed above are not academic exercises. They feed many decisions about contracts, trades, and roster construction, which means the ethical implications of how they are built and used matter as much as their statistical accuracy. The standard reference for thinking about these implications is Barocas, Hardt, and Narayanan's *Fairness and Machine Learning: Limitations and Opportunities* (Barocas, Hardt, & Narayanan, 2023), which lays out the major frameworks for evaluating algorithmic fairness including group-level fairness (parity of outcomes across demographic or role groups), individual fairness (similar individuals receiving similar predictions), and the trade-offs between them. Applied to NBA player evaluation, these frameworks raise concerns that the shot-quality literature has not always engaged with directly. A model that ranks players purely by shot-quality residual may systematically disadvantage players whose roles force them into low-expected-value shots — primary creators carrying heavy usage, or bigs setting screens in physical interior situations — even if those players are performing well within their assigned role. By building a shot-quality model that addresses these issues, we can develop a model that may well be used in NBA analytics themselves.

Where Our Project Fits

Our work builds directly on the EPV and shot-quality models rooted in the Cervone et al. (2014) papers, while drawing on the residualized-evaluation methodology of Bajons and Kook (2025) and the modern Bayesian approach of EPAA (2024). The three research questions we proposed map well onto these threads: RQ1, which identifies players and teams that consistently outperform their expected shot value, is a direct application of residualized player evaluation to NBA shot data; RQ2, which traces how shot success by court location has changed across the tracking era, extends the longitudinal view of the analytics era that Poropudas and Halme (2023) describe at the team level; and RQ3, which compares shot-quality-adjusted rankings against traditional efficiency metrics, connects our work to the broader question of whether the shot-quality lens tells a different story than the box-score-derived metrics that still dominate NBA discourse. The ethics framing from Barocas et al. (2023) runs through all three questions, particularly in how we interpret which players and roles the model rewards or penalizes.

References

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